



AGE AND GROWTH OF THE COMMERCIAL SPINY LOBSTERS *PANULIRUS ARGUS* (LATREILLE, 1804) AND *PANULIRUS LAEVICAUDA* (LATREILLE, 1817) (DECAPODA, PALINURIDAE) IN NORTHEAST BRAZIL: A REVIEW

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ABSTRACT

The two spiny lobster fisheries targeting *Panulirus argus* (Latreille, 1804) and *Panulirus laevicauda* (Latreille, 1817) have made an important marine resource in Northeast Brazil since 1955. The Von Bertalanffy age and growth parameters of the spiny lobster started being studied in 1960 and the last assessment was performed over 20 years ago. The objective of this review is to analyse the type of input data and the methods historically used to assess age and growth of the spiny lobster stock in Northeast Brazil.

Our study reviews and updates the research on the subject and proposes relative values for the Von Bertalanffy growth parameters for *P. argus* and *P. laevicauda* as provisional reference. We recommend further age and growth research on spiny lobsters to create a new and updated growth function for commercial lobsters in Northeast Brazil.

Key words. — *Panulirus argus*, *Panulirus laevicauda*, spiny lobster fishery, age and growth, Northeast Brazil

RESUMO

A pesca de lagosta de duas espécies-alvo, *Panulirus argus* (Latreille, 1804) e *Panulirus laevicauda* (Latreille, 1817) tem sido um importante recurso marinho no Nordeste de Brasil desde 1955. Em aquela região os parâmetros de idade e crescimento da função de Von Bertalanffy começaram a ser estudados desde o ano 1960 e por última vez há mais de 20 anos. O objetivo deste estudo de revisão é analisar os dados de entrada e métodos antigamente usados para assessorar a idade e crescimento do estoque de lagostas espinhosas explotadas no Nordeste do Brasil. Este trabalho atualiza o estado do conhecimento do tema e propõe valores relativos para os parâmetros da função de crescimento de Von Bertalanffy para *P. argus* e *P. laevicauda* como referência provisional. Sugerimos continuar com a pesquisa focada na idade e crescimento das lagostas espinhosas para poder facilitar

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a geração de uma nova e atualizada função de crescimento para as lagostas com valor comercial no Nordeste do Brasil.

Palavras-chave. — *Panulirus argus*, *Panulirus laeviscauda*, pescarias de lagosta, Nordeste do Brasil

INTRODUCTION

Growth in species of Crustacea has been studied over the last decades and the results indicate a complex process that involves a series of several moulting events (ecdysis) as well as intermoult (diecdysis) stages, necessary for an increase in the body size of a given organism during its lifetime (Hartnoll, 1983). As a result, crustacean age and growth data seem to fit better a non-continuous, stepped trend-line rather than a classical curve, that fits the majority of, e.g., true fish ("Pisces") growth data. The rigid exoskeleton of crustaceans simply leaves no other option. Crustacean growth can be influenced by several factors, including water temperature (Cruz et al., 1981), food availability (Phillips et al., 1992), habitat productivity (Morgan, 1985), reproduction (Hartnoll, 1985), disease (O'Brien & Van Wyk, 1985), and pollution (Fingerman, 1985).

Age and growth assessments of crustacean species are a strong tool for fisheries management to set the status of an exploited stock in function of lifespan, first capture, age at-first-maturity, age at-minimum-landing-size, cohort identification and population dynamics (Beverton & Holt, 1957; Ehrhardt, 2008), among others. Worldwide, marine and freshwater crustacean fisheries produce more than 5×10^6 metric tons (FAO, 2006). Within crustacean fisheries, lobster production worldwide reached 214 240 metric tons in 1993, composed of 37.9% *Homarus americanus* H. Milne Edwards, 1837 and 19.2% *Panulirus argus* (Latreille, 1804) (FAO, 1993). Moreover, there has been an international effort on age and growth assessment for *P. argus* in more than ten countries across the Western Central Atlantic Region (FAO, 2001). Also, different methodologies have been tested in the Great Caribbean to assess the age and growth of *P. argus*, including: (1) tagging and recapture in Florida U.S.A., Cuba, the U.S. Virgin Islands, and the Bahamas (Little, 1972; Olsen & Klobic, 1975; Waught, 1981; Phillips et al., 1992; FAO, 2001; Ehrhardt, 2008); (2) the Bhattacharya method in Cuba (Cruz et al., 1981); and (3) the Fabens method in Bahia de la Asunción, Mexico (Lozano-Alvarez et al., 1991).

Furthermore, in Northeast Brazil (NEB) the lobster stock composed of the two target species, *Panulirus argus* and *P. laeviscauda* (Latreille, 1817), has been an important fishing resource since 1955 (Fonteles-Filho et al., 1988; Fonteles-Filho, 1992). The export of these two lobster fisheries in NEB between 1955 and 1991 constituted approximately 750 billion US\$ (Fonteles-Filho, 1994). The age

and growth of *P. argus* was assessed for the first time in 1964 by Dos Santos et al. (1964), using fisheries-dependent data, total length frequency distribution, and the Von Bertalanffy growth function (VBGF). Subsequently, other important studies were performed using similar input data and methods. Age and growth assessments of *P. laeviscauda* were performed for the first time in 1973 by Dos Santos & Ivo, using total length frequencies and VBGF, based on fisheries-dependent data.

Some of the limitations of the previous age and growth assessments seems to be related to the too few sources of raw data and the sampling procedures involved. The last age and growth assessment of *P. argus* and *P. laeviscauda* in NEB was over 20 years ago (Ivo & Pereira, 1996), and still there are no accurate answers to basic questions like: “At what age do the lobsters reach their first maturity?” and “At what age do the lobsters reach minimum landing size?”

The objective of this review is to analyse the previous age and growth assessments used for *P. argus* and *P. laeviscauda* in NEB since 1962. To do so, we collected available publications on the subject to compare and evaluate previous age and growth assessment results as claimed by several authors. We, also propose relative Von Bertalanffy growth parameters as provisional data until new growth parameters will be developed by fisheries science in NEB. We focus our discussion on the raw input data and methods used in previous assessments and propose some recommendations.

We believe there are compelling reasons to re-analyse and continue age and growth research and assessments of the commercial lobsters *P. argus* and *P. laeviscauda* in NEB. The first one is the fact that we are dealing with a two-target-species lobster fishery, which is a special characteristic that is less common in other areas of the Western Atlantic Region. Another important feature is that it has been hypothesized that self-recruitment occurs in the *P. argus* stock on the Brazilian shelf (Cruz et al., 2015) and this might indicate different growth trends than those found in other areas of the species’ distribution. The last reason is, to gain a better understanding of spiny lobster growth trends in relation to high exploitation with non-selective fishing gear in the last 50 years. We believe that research focused on the palinurid lobster stock in NEB will expand knowledge of fisheries science in the region of the Western Atlantic.

METHODS

Area of study

Spiny lobsters are distributed throughout the Brazilian continental shelf from Amapá state (04°26’N 51°32’W) to Espírito Santo state (21°17’S 40°56’W) (Cruz

et al., 2013). For our review project we consider Northeast Brazil (NEB) as the area between 2°30'N to 18°S and 35°W to 47°W (Cruz & Bertelsen, 2009).

Length data standardization

The variable total length (TL) in centimetres was normally used to perform previous spiny lobsters age and growth assessments in NEB. In this review, the total length (TL) in centimetres was converted into cephalothorax length (CL) in millimetres using a logarithmic equation for each species and for each sex separately, as proposed by Rocha & Xavier (2000). For *Panulirus argus*:

- (1) males: $\ln Y = 1.61 + 0.86 \ln X$
- (2) females: $\ln Y = 1.29 + 0.94 \ln X$

Likewise for *Panulirus laeviscauda*:

- (3) males: $\ln Y = 1.73 + 0.82 \ln X$
- (4) and females: $\ln Y = 1.33 + 0.93 \ln X$

Previous growth assessments for spiny lobsters in NEB

Input data and methods used for Von Bertalanffy growth parameters

A bibliographic compilation was made in order to gather all the information on spiny lobsters' age and growth previously assessed in the NEB region since 1960. The input data and methods to perform Von Bertalanffy growth function analysis (VBGF; Von Bertalanffy, 1938) used by those previous assessments were examined and chronologically compared for both species of commercial spiny lobster, i.e., *Panulirus argus* and *Panulirus laeviscauda*.

Evaluating the input data and methods for the Von Bertalanffy growth function

The spiny lobster input data and the methods for VBGF used previously in NEB were evaluated by comparing with other growth estimations of *Panulirus argus* in the Great Caribbean. A total of five criteria for input data developed in the Great Caribbean by different authors were identified to set up a comparative evaluation (table I). Criteria were selected based on the kind of data used and the relevance of the studies at that time (Morgan, 1980; Cruz, et al., 1981). The same procedures were applied for *Panulirus laeviscauda* due to fact that the age and growth of that species have been studied only in NEB and no publications on the subject were found from other areas of the Western Atlantic Region.

TABLE I

Criteria to evaluate the input data and methods used in previous age and growth assessments for spiny lobsters in the Northeast Brazil area

No.	Criteria	Author
1	Growth information of spiny lobsters retained in laboratory conditions	Sutcliffe (1957)
2	Planktonic age information included	Lyons et al. (1981)
3	Von Bertalanffy growth function adjustments (Beverton & Holt, 1957; Ford-Walford: cf. Ford, 1933 and Walford, 1946)	Cruz et al. (1981)
4	Tag and recapture information included	Peacock (1974); Phillips et al. (1992)
5	Length frequency distribution range from <25 to >125 mm CL	Phillips et al. (1992)

Comparing and evaluating output data of the Von Bertalanffy growth function
The Von Bertalanffy growth function (VBGF):

$$L_t = L_{\infty}(1 - \exp(-K(t - t_0)))$$

(Von Bertalanffy, 1938) was used to graphically compare trends between different, previous age and growth assessments for spiny lobsters, using an Excel spreadsheet. To evaluate the output performance, values of asymptotic length (L_{∞}) were compared with maximum length ($L_{\max}$) observed for spiny lobster landings in the NEB area as implemented by Cruz and collaborators for the southwestern Cuban shelf (Cruz et al., 1981).

Current growth estimations for spiny lobster in NEB: calculating the relative
Von Bertalanffy growth parameters L_{∞} and K

The relative Von Bertalanffy growth parameters: (1) asymptotic length (L_{∞}), and (2) growth coefficient (K), for the spiny lobster result from the calculated mean values of previous age and growth assessments reviewed in this study. The relative Von Bertalanffy parameters were calculated by species, by sex, and graphically presented with a 95% confidence interval as implemented before for *Panulirus argus* in other regions (Phillips, 1992). The relative values for *P. argus* and *Panulirus laeviscauda* are here proposed as provisional reference data for spiny lobsters in the NEB area until values from an updated model can take their place.

RESULTS

Length data standardization

Since the beginning of spiny lobster fisheries research in the year 1961, the variable total length (TL) in centimetres was observed as “the distance between the

notch formed by the rostral spines and the posterior part of the telson” (Pinto Paiva & da Silva, 1962). To perform growth assessments in NEB, some authors in those previous assessments transformed total length into cephalothorax length using the same, single linear equation for both sexes and both species. The logarithmic equations used by us in this review to convert the total length into the cephalothorax length separately by species and by sex, showed a good fit when the output in the form of growth curves was graphically compared.

Previous growth assessments for spiny lobsters in NEB

Input data and methods previously used for Von Bertalanffy parameters

The results show that from the year 1964 up to 2008, four age and growth assessments were performed for *Panulirus argus* and two such assessments for *Panulirus laeviscauda*. These previous assessments collected different body measurements, like total length and tail length, as input data for both species (table II). The methods used were based on length frequency analysis purported to perform

TABLE II

Source of the studies, input data, and methods used to perform the Von Bertalanffy growth function analysis for the spiny lobsters *Panulirus argus* (Latreille, 1804) and *Panulirus laeviscauda* (Latreille, 1817) in Northeast Brazil from 1964 to 2008

Author	Measuring growth	Method	Type of data
<i>Panulirus argus</i> (Latreille, 1804)			
Dos Santos et al. (1964)	Total length frequencies	Modal Progression Analysis	Fisheries-dependent data
Gonzalez-Cano & Rocha (1995)	Total length frequencies	Shepard's Length Composition Analysis	
Ivo & Pereira (1996)	Total length frequencies (including tail length*)	Electronic Length Frequency Analysis (ELEFAN, Bhattacharya's method)	
Leocádio & Cruz (2008)	Cephalothorax length frequencies	Meta-analysis	
<i>Panulirus laeviscauda</i> (Latreille, 1817)			
Dos Santos & Ivo (1973)	Total length frequencies	Modal Progression Analysis	Fisheries-dependent data
Ivo & Pereira (1996)	Total length frequencies (including tail length*)	Electronic Length Frequency Analysis (ELEFAN, Bhattacharya's method)	

*Total length and tail length data were converted to cephalothorax length.

the Von Bertalanffy growth function calculations. Chronologically, the methods began with basic progression analysis in 1964 to turn to computational programs in 1996. None of the previous assessments stated why VBGF was used. Fisheries-dependent data were the only information used for both species (table II), but the reason for using such a source was not explained by any of the authors.

Evaluating the input data for the Von Bertalanffy growth function

The five criteria selected were adequate to highlight the relevance of using raw input data in previous age and growth assessments in NEB. The input data evaluations for the VBGF in NEB for spiny lobsters show that only one criterion (criterion No. 3, which refers to procedures for calculating the Von Bertalanffy growth parameters known as the Beverton & Holt adjustment (cf. Beverton & Holt, 1957), and the Ford-Walford linear method (cf. Ford, 1933; Walford, 1946; Gulland, 1969; Ricker, 1975; Vaughan & Kanciruk, 1982)), matched with other studies developed in the Great Caribbean in a similar time frame (table III). The criteria selected show that all previous age and growth assessments in NEB were run with limited input data. For example, criterion number five (length frequency

TABLE III
Input data and methods used in previous age and growth assessments for the spiny lobsters *Panulirus argus* (Latreille, 1804) and *Panulirus laeviscauda* (Latreille, 1817) in Northeast Brazil and comparison with other such assessments in the Great Caribbean

Author	Method		Input data		
	Beverton & Holt, 1957; Ford-Walford	Length frequency range from <25 to >125 mm CL	Laboratory reared information	Planktonic age information	Tag and recapture information
<i>Panulirus argus</i> (Latreille, 1804)					
Dos Santos et al. (1964)	Ford-Walford	≥63-132 mm CL	No input data		
Gonzalez-Cano & Rocha (1995)	Not available	≥50-122 mm CL			
Ivo & Pereira (1996)	Ford-Walford	Not available			
Leocádio & Cruz (2008)	Not available	≥50-132 mm CL			
<i>Panulirus laeviscauda</i> (Latreille, 1817)					
Dos Santos & Ivo (1973)	Ford-Walford	≥46-113 mm CL	No input data		
Ivo & Pereira (1996)	Ford- Walford	Not available			

data) indicates that the ranges of length input data are limited to fisheries-dependent values between 50 and 132 mm CL, with no values below 50 mm CL for *Panuliurs argus*, and from 46 to 113 mm CL with no values below 46 mm CL for *Panulirus laevicauda* (table III). Also, previous growth assessments for both species did not give raw input data regarding laboratory reared spiny lobsters, planktonic age of spiny lobsters, or tag & recapture information, for performing VBGF. There is no clear explanation why the assessments at that time were limited to only the Ford-Walford linear method, and did not include values below 25 mm CL.

Comparing and evaluating previous output data of Von Bertalanffy growth parameters for spiny lobsters in NEB

The first *Panulirus argus* age and growth assessments were performed by Dos Santos and collaborators (Dos Santos et al., 1964), using length frequency to adjust VBGF. The authors calculated the growth coefficient “*K*” with the highest values for males and females found for this review (table IV). This assessment highlights the similarity in growth between males and females that began in the first year and lasted until the third year, when males started showing a different trend that becomes more pronounced in the fifth year (fig. 1A). This growth similarity between sexes during the first years of the species’ life was not explained by the authors. The evaluation of the output shows a maximum length ($L_{\max}$) > asymptotic length (L_{∞}) scenario for *P. argus* males and females. Wherein the Von Bertalanffy growth parameter L_{∞} does not represent the observation of $L_{\max}$ found in the spiny lobster landings. The $L_{\max}$ observed in spiny lobster landings in the NEB area for males was reported to range from 141 to 146 mm CL (Pinto Paiva & Da Costa, 1970; Do Nascimento et al., 1984) and for females from 125 to 135 mm CL (Pinto Paiva & Da Costa, 1966, 1967, 1968; Do Nascimento, et al., 1984).

TABLE IV

Output of Von Bertalanffy growth parameters from previous assessments of *Panulirus argus* (Latreille, 1804) in Northeast Brazil; the Dos Santos et al. (1964) model is the only one showing an $L_{\max}$ greater than L_{∞} scenario

Author	L_{∞}		K		Output data evaluation ($L_{\max}$ vs. L_{∞})
	♂	♀	♂	♀	
Dos Santos et al. (1964)	142.0	130.1	0.340	0.380	$L_{\max} > L_{\infty}$
Gonzalez-Cano & Rocha (1995)	207.2	124.9	0.260	0.230	$L_{\max} < L_{\infty}$
Ivo & Pereira (1996)	178.4	160.1	0.229	0.236	
Leocádio & Cruz (2008)	176.7	142.4	0.279	0.281	

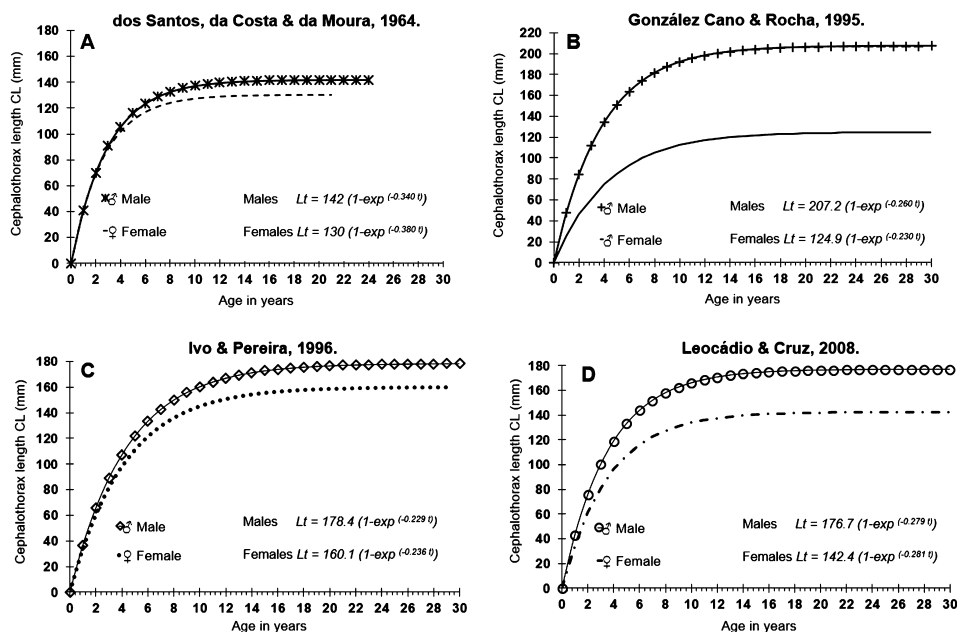


Fig. 1. Output results of the Von Bertalanffy growth function for *Panulirus argus* (Latreille, 1804) in Northeast Brazil. A, Dos Santos et al., 1964; B, González-Cano & Rocha, 1995; C, Ivo & Pereira, 1996; D, Leocádio & Cruz, 2008. A and C show different growth trends between males and females after approximately three years. B and D show different growth trends for males and females almost at their first year. Decreasing order of asymptotic length (L_∞) ♂, B; C; D; A. ♀, C; D; A; B. Decreasing order of growth coefficient (K) ♂, A; D; B; C. ♀, A; D; C; B.

The age and growth assessment of Gonzalez-Cano & Rocha (1995) shows, graphically, considerable differences compared to the previous model (fig. 1B). The figures show notable differences in growth trends for males and females after the first year, that become more pronounced in the second year.

In 1996, Ivo & Pereira performed a new growth curve analysis using length frequencies, the Batacharraya method, and ELEFAN I (see Gayanilo, Jr. et al., 1989; Pauly & Gayanilo, Jr., 1990) for VBGF for males and females of *P. argus* (fig. 1C). Compared with the previous model, the Ivo & Pereira (1996) model displays a difference between males and females starting at two years, that becomes more pronounced at three years. In this assessment, the authors claim a non-growth-rate difference between males and females of the species, but they did not explain the methodological procedures to calculate that rate between sexes.

A new growth assessment was performed in 2008 by Leocádio & Cruz based on a meta-analysis graphically showing a trend different from the preceding model (fig. 1D). This model illustrates different growth trends between males and females after the first year. The evaluations of the last three assessments described above, highlight a $L_{\max} < L_\infty$ scenario for males and females of *P. argus*.

TABLE V

Output of Von Bertalanffy growth parameters from previous assessments of *Panulirus laeviscauda* (Latreille, 1817) in Northeast Brazil

Author	L_{∞}		K		Output data evaluation ($L_{\max}$ vs. L_{∞})
	♂	♀	♂	♀	
Dos Santos & Ivo (1973)	83.3	83.3	0.708	0.708	$L_{\max} > L_{\infty}$
Ivo & Pereira (1996)	190.12	153.0	0.221	0.226	$L_{\max} < L_{\infty}$

Wherein potentially the Von Bertalanffy parameter L_{∞} might be overestimated when compared with the $L_{\max}$ observations of the species landings (table IV).

On the other hand, the first study to assess the age and growth of *Panulirus laeviscauda* was performed by Dos Santos & Ivo (1973). In this work the authors used length frequency distribution and the Ford-Walford linear method for performing VBGF (table V). In this assessment, the authors suggest the non-growth difference hypothesis between males and females for the species mentioned. The premises of the non-growth hypothesis were, however, not fully explained by the authors. Graphically, this model shows little growth development after 5 years for both sexes (fig. 2). The output evaluation of the assessment indicates a $L_{\max} > L_{\infty}$ scenario, where the asymptotic length L_{∞} (83.4 mm CL) is smaller than the $L_{\max}$ observations of the *P. laeviscauda* landings, which were reported in a range from 108 to 124 mm CL for males and 94 to 108 mm CL for females in the NEB area (Pinto Paiva & Da Costa, 1965; Da Costa & Lima Paiva, 1974; Do Nascimento et al., 1984) (table V). Age and growth of *P. laeviscauda* were assessed again by Ivo & Pereira (1996). This assessment graphically shows a Von Bertalanffy growth curve different between males and females after the first year, the difference becoming more pronounced at five years (fig. 2). The conclusion proposed that there was no difference in growth rate between males and females of the species. Ivo & Pereira (1996) did not explain the methodical procedures to calculate the growth rate for males and females of *P. laeviscauda*.

Current calculations: relative values for Von Bertalanffy growth parameters L_{∞} and K for spiny lobsters in NEB

The relative values from previous VBGF parameters for spiny lobsters show differences between males and females for both species. *Panulirus argus* males present higher L_{∞} and lower K relative values when compared with the females (table VI). Graphically, the species mentioned show similar growth trends for males and females from the first year, but become different after the second year.

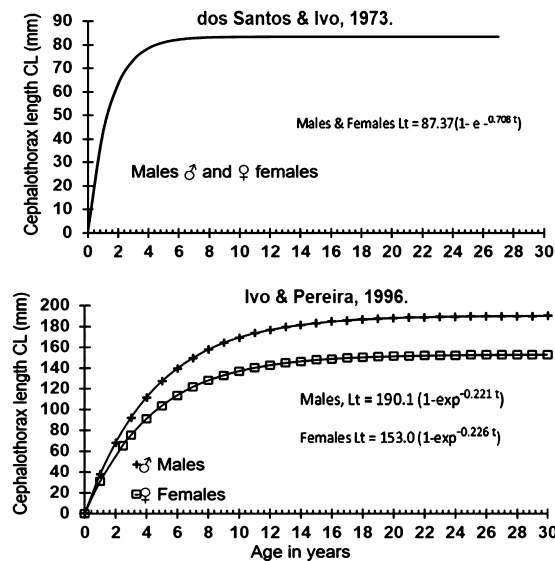


Fig. 2. Output results of the Von Bertalanffy growth function for *Panulirus laeivicauda* (Latreille, 1817) males and females in Northeast Brazil. A, Dos Santos & Ivo, 1973, for males and females together, showing high growth during the first three years, with $L_{\infty} = 83.3$ mm CL and $K = 0.708$; B, Ivo & Pereira, 1996, showing different growth for males and females of the species after the second year. Males, $L_{\infty} = 190.1$ mm CL and $K = 0.221$. Females, $L_{\infty} = 153.0$ mm CL and $K = 0.226$.

This difference is notable after the third year (fig. 3). Relative values show that males and females from the mentioned species reach the first year measuring 41.9 mm and 33.8 mm CL, respectively (table VI). Likewise, *Panulirus laeivicauda* males present higher L_{∞} and lower K values compared with the females of that species (table VI). Graphically, males and females show the same growth rate until

TABLE VI
Output of relative Von Bertalanffy growth parameters for *Panulirus argus* (Latreille, 1804) and *Panulirus laeivicauda* (Latreille, 1817) in the Northeast Brazil marine area; relative values are proposed as provisory

This study; proposal	L_{∞}		K		Cephalothorax length (mm) at 1 year old	
	♂	♀	♂	♀	♂	♀
<i>Panulirus argus</i> (Latreille, 1804)						
Relative value	176.1	139.4	0.277	0.282	41.9	33.8
<i>Panulirus laeivicauda</i> (Latreille, 1817)						
Relative value	136.7	118.2	0.460	0.490	40.0	36.6

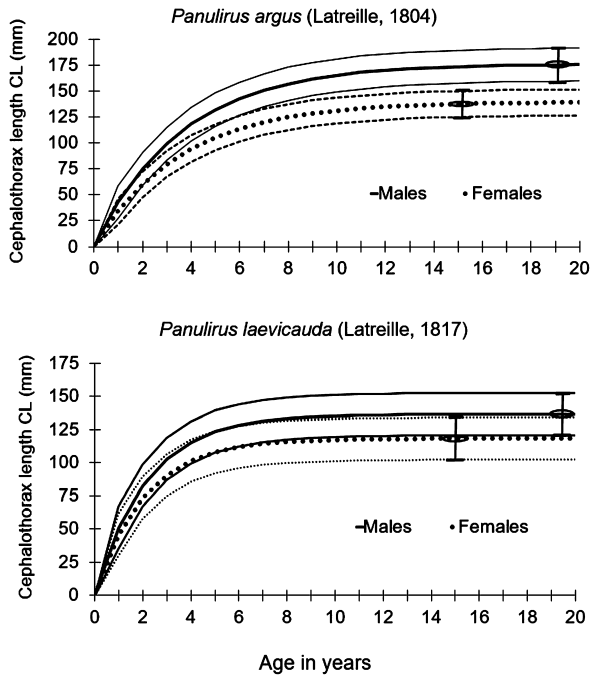


Fig. 3. Relative output results of the Von Bertalanffy growth function (95% confidence interval) for *Panulirus argus* (Latreille, 1804) and *Panulirus laevicauda* (Latreille, 1817) in Northeast Brazil. ♂, solid line; ♀, dotted line.

the third year, when males show a higher growth trend (fig. 3). Males and females reach the first year measuring 40.0 and 36.6 mm CL, respectively.

DISCUSSION

Spiny lobster body measurements and length data standardization

Previous spiny lobster age and growth assessments converted different spiny lobster body measurements (lengths) for analysis. This was especially noted when long time series (decades) with different body measurements were transformed and joined together to create a standard data set. Some data sets included conversions of tail length to total length. For example, in Ivo & Pereira (1996) tail length was transformed to total length without reporting which formula was used for such morphometric transformation. It is important to mention that the logarithmic equations developed by Rolim & Rocha (2000) and used by us in this review showed a good adjustment for the standardization of the data by species and sexes. Even so, we consider that the different methodologies previously used to transform spiny lobster measurement in NEB created low quality length frequency

distribution data and thereby a potential source of bias in the growth assessments. The observation of cephalothorax length as “notch of the supra orbital spines until the posterior part of the cephalothorax” (Cruz et al., 2011) has been considered the conventional index of body size for growth studies (Wahle & Fogarty, 2006). Observation of lobster cephalothorax length is a more accurate variable than total length, due to the flexibility of the tail during measurement. The use of linear equations to transform the tail length into total length might also be a source of bias in *Panulirus argus* research (Cruz & Bertelsen, 2009; Cruz et al., 2011).

Previous studies

Methods and raw input data used for Von Bertalanffy growth parameters and their evaluations

Previous age and growth assessments of the spiny lobsters *Panulirus argus* and *Panulirus laeviscauda* in NEB used limited input data and methods when compared with other studies developed in the Great Caribbean in a similar time frame. There is no evidence of data input from laboratory reared, tag & recapture, or planktonic age material. Only fisheries-dependent data were employed to determine length frequency distributions for the Von Bertalanffy growth function analysis. Length frequency measurements can be obtained cost-effectively from fish landing places and have been used in various lobster taxa (Pauly & Morgan, 1987; Wahle & Fogarty, 2006). These data provide a medium level of biological reality for describing the average growth of individuals, but have a low relevance for the individual (Chang et al., 2012). Since all length frequencies used in previous assessments come from fisheries-dependent data it is important to highlight two crucial things. The first is that different types of fishing gear normally used in commercial fishing can be a source of bias in the sampled spiny lobsters, and accordingly in the length frequency distributions, as well as in the age and growth calculations. It seems as if previous growth assessments did not consider these facts as a possible source of bias for growth analysis purposes. The second one comprises, that apparently length frequency distributions used as input in previous assessments were not designed for purposes of growth analysis. Is not clear if the length frequency used as raw input is part of some experimental design. The sample sizes, sex ratios, random sample selection, small or large spiny lobster proportion in the catch, and the externally observable information on the reproductive state of the females, were not reported. Also, the length distribution range is limited from 50 to 132 mm CL for *P. argus* and from 46 to 113 mm CL for *P. laeviscauda*, with no observations available below 25 mm CL for each of the species. Observations below 25 mm CL have been described as important input data for age and growth estimations in *P. argus* (cf. Phillips, 1992).

On the other hand, previous spiny lobster age and growth assessments in NEB implemented the Von Bertalanffy growth function. None of the authors of the reviewed assessments explained clearly why VBGF was selected to perform the assessment. Also, there is no reference if other age and growth models have been used in NEB at that time. VBGF has been widely used to assess fish growth, and has also been used to assess crustacean growth (Caddy, 1986; Cobb & Caddy, 1989). More than nine species of lobsters, including *P. argus*, have been assessed with VBGF in different regions worldwide (Wahle & Fogarty, 2006). Some of the problems to assess the age and growth of spiny lobsters with a continuous growth model like VBGF are, that the model does not capture discontinuous growth due to the periodic moulting process. In contrast, satisfactory outcomes of VBGF analyses result when proper data are used as input. For example, tag and recovery observations and measurements of intermoult increments, and their probabilities. Unfortunately, all this is not the case for the previously performed age and growth assessments of spiny lobsters in NEB.

From another perspective, for growth assessment it is important to have a good knowledge of the cephalothorax-length-to-body-length morphometrics of the spiny lobster in order to acquire reliable growth measurements. This kind of information is limited for *P. argus* and *P. laeviscauda* in NEB. Finally, the scenario of previous age and growth assessments for spiny lobsters in NEB can be summarized as consisting of low statistical-power length frequency distributions from fisheries-dependent data, used as input in a model (VBGF) that assumes continuous growth. With this scenario, we considered that the age and growth values stated by previous assessments in NEB are doubtful, and that there obviously is a need for a new assessment.

Comparing and evaluating output data of Von Bertalanffy growth parameters for spiny lobsters in NEB

Graphically, VBGF output indicates considerable differences between the previous assessments for *Panulirus argus* in NEB. As general observation we can divide these differences into two trends. For one side, the first trend is the one with similar growth between sexes during the first three years of age. On the other side, the second one has different growth trends between sexes during the first years. Also, the $L_{\max}$ for *P. argus* indicates two different scenarios for the model outputs where the asymptotic length L_{∞} is either less than, or more than the $L_{\max}$, and with a poor representation of the observations from the species' landings.

On the other hand, outcomes of age and growth assessments for *Panulirus laeviscauda* are limited due to the fact that the male and female values have been included in the same model. Also, the $L_{\max}$ for the species is greater than the calculated L_{∞} , and K seems to be overestimated. The Ivo & Pereira (1996)

model shows a difference between males and females from the 2nd year onward, but L_{∞} seems seriously overestimated when compared with the $L_{\max}$ of the landings observations. This scenario takes us to ask the following question for both species: “If the raw input data and methodological conditions between the models (VBGF) are similar, why do we find these differences in the output?” We believe that rather than biological or environmental factors influencing the growth of the species, the differences between the models’ outcomes are due to the type of input data, that concern too low quality length frequency distributions for *P. argus* and *P. laeviscauda*. Sample sizes and the proportions of small and large spiny lobsters in the samples corrupt the purity of the analyses, as was discussed in the previous section. Also, it seems as if the conversions of the various body measurements (e.g., total length and tail length) might result in the computation of considerable differences in the output. Consequently, we strongly believe that the VBGF parameters L_{∞} and K must be recalculated with proper input data, and such for both species here at issue.

Current calculations: relative Von Bertalanffy growth parameters L_{∞} and K for spiny lobsters in NEB

The difference between sexes of the VBGF parameters calculated by us for *Panulirus argus* have also been reported for that species in the Great Caribbean. The L_{∞} calculated by us for NEB males is lower than the values reported for males in the Great Caribbean. In the same sense, the females in the NEB area had a lower L_{∞} compared with the females in the larger region mentioned (Leocádio & Cruz, 2008).

Likewise, the K figures for males in NEB are higher than those of *P. argus* males in the Great Caribbean, and the same holds true for the females: those in NEB showed higher K values than the ones reported for specimens from the Great Caribbean (Leocádio & Cruz, 2008). In the Cuban archipelago, the most likely values of K were reported within a range of 0.200 to 0.380 for *P. argus* (cf. Cruz et al., 1981). On the other hand, differences in the growth parameters L_{∞} and K between *P. argus* males and females have been described for Florida, U.S.A. (Little, 1972), the U.S. Virgin Islands (Olsen & Klobick, 1975; Mateo & Tobias, 2002), and the Bahamas (Waugh, 1981). Moreover, our VBGF age calculations for *P. argus* males show that they reach the first year measuring a mean of 41.0 mm CL and that females do so with 33.8 mm CL on average. However, different results have been found for *P. argus* in the Cuban archipelago, where the species reaches the first year measuring an average of 10 mm CL (Baisre, 1979), and reaches the second year measuring an average of 63 mm CL (Cruz et al., 1981). We believe that our calculated age-at-length is overestimated, and thus needs a new assessment, based on proper input data.

As regards *Panulirus laevicauda*, that discussion does not seem to be an optimistic one. To our concern, age and growth assessments for *P. laevicauda* outside NEB and, in general, for other areas of the Western Atlantic Region, are nonexistent. This fact leaves a difficult scenario, with no information to compare. We consider, though, that our calculations can function as a stepstone for future research.

In summary, we think we can conclude that our review, rather than answering all questions about age and growth of these two commercial spiny lobsters, brings an updated perspective on the subject. We believe that the relative values for the VBGF parameters as calculated herein, can act as a starting point to enter a new era of research into age and growth of *P. argus* and *P. laevicauda* in NEB. We are proposing relative values of the Von Bertalanffy growth parameters for species and sexes as follows:

***Panulirus argus*:**

♂ Males, $L_{\infty} = 176.1$, $K = 0.277$, reaching the first year measuring 41.9 mm CL

♀ Females, $L_{\infty} = 139.4$, $K = 0.282$, reaching the first year measuring 33.8 mm CL

***Panulirus laevicauda*:**

♂ Males, $L_{\infty} = 136.7$, $K = 0.460$, reaching the first year measuring 40.0 mm CL

♀ Females, $L_{\infty} = 118.2$, $K = 0.490$, reaching the first year measuring 36.6 mm CL

We consider these relative age and growth values as provisory, under the precautionary approach to fisheries research (FAO, 1996), until a new growth parameter investigation will take place by all fisheries-science stakeholders in Northeast Brazil.

MANAGEMENT IMPLICATIONS

As a consequence of our above results, we recommend the development of a new age and growth assessment for *Panulirus argus* and *Panulirus laevicauda*, with the following recommendations:

Input data for age and growth assessment

An important lesson learned, based on the deficiency of the previous age and growth assessment on the NEB spiny lobster fauna is: low quality input data should be avoided. That goal can be reached as follows:

(1) **Length frequency distribution data.**— These should come from an excellent long-term fisheries-independent data-sampling design (Ricker, 1975; Pauly & Morgan, 1987; Fonteles-Filho & Holanda, 1990; Cruz et al., 2011). This should include, but not be limited to: defining the number of samples per unit

time, registration of the sex ratio, standardizing gear selectivity, and observations of cephalothorax length over a range of <25 mm to >125 mm CL (Phillips et al., 1992).

(2) **Tag and recapture.**— It is important to set a long-term tag and recapture program for the spiny lobster in NEB. A successful tag and recapture program should consider but not be limited to: be planned for five years or more, at least one thousand organisms should be tagged, double tagging should be considered to minimize total tag-loss, recapture information for males and females should be collected separately, and the time between tag and recapture should be planned in advance so as to be not less than 100 days. Interesting suggestions resulting from the tagging of *Panulirus argus* and other lobster species have been brought forward by several authors already (Cruz et al., 1986; Phillips et al., 1992; Wahle & Fogarty, 2006; Ehrhardt, 2008).

(3) **Spiny lobster morphometrics.**— We believe that for a successful growth assessment it is very important to update the information on the morphometric relationships by sex and by species in order to better understand the development of the growth of the body of spiny lobsters. Interesting and easy to replicate methods for the assessment of morphometrics have been proposed by Rolim & Rocha (1972), Do Nascimento et al. (1984), and Rocha & Xavier (2000).

(4) **Spiny lobster laboratory rearing.**— Rearing under laboratory conditions will generate important input data for small spiny lobsters (<25 mm CL) and thus provide information about the pre- and post-moult stages in the ecdysis cycle, of which little is known as yet for the populations in NEB. Important methods for the observation of ecdysis in *P. argus* were proposed by Travis (1954) and for laboratory rearing by Matthews & Maxwell (2007) and Dahlgren & Staine (2007). A regional methodology was developed by Assad and collaborators, who experimented with marine nursery cages for raising juveniles of *Panulirus argus* and *P. laevicauda* in NEB (Assad et al., 1992).

Modelling age and growth of spiny lobsters in NEB

With the construction of robust input data there will be a need to select “one perfect” numerical model to assess the age and growth of spiny lobsters. However, it seems as if we can not keep thinking that way. A new way of thinking must propose the use of different types of numerical models to expand the growth of an individual spiny lobster to the average growth of a stock, according to biological reality. The following list gives an idea of the different models that potentially can be used:

(1) **Continuous model Von Bertalanffy.**— VBGF will fit mainly raw input data from: (a) tag and recapture data; and (b) moult increment information (Wahle & Fogarty, 2006).

(2) **Continuous model Von Bertalanffy-Fabens-recapture.**— This model takes as input information from recaptured lobsters and the delay-in-time recapture, and has been used with *Panulirus argus* (cf. Campell & Phillips, 1972; Lozano-Alvarez et al., 1991; Phillips et al., 1992).

(3) **Intermoult period.**— This model was used with *P. argus* data from tag & recapture and from laboratory reared specimens (Ehrhardt, 2008).

(4) **Modern ELEFAN-based fitting method.**— This recently developed method seems to find good adjustment with data-poor scenarios. The model uses length frequency analysis for fitting a curve with several algorithms (Schwamborn et al., 2019).

CONCLUSIONS

From the data source approach as herein applied, our findings are highlighting that the type of raw data used as input is as important as the numerical model for calculating realistic simulations of spiny lobster age and growth. Earlier studies in the NEB area are characterized by limitations in the source data and in the sampling efforts, but can still be considered an important biological baseline for spiny lobster research. This goes especially for *Panulirus laevicauda*, which has been little studied in the Western Atlantic Region until now. We used the results of those previous studies to calculate relative age and growth values for *Panulirus argus* and *Panulirus laevicauda*, but the outcomes have to be used with caution. We strongly believe that our recommendations, especially the creation of a database of long-term fisheries-independent input data, would enhance a framework for future age and growth modelling for *P. argus* and *P. laevicauda*, according to the biological reality of the stock in the Western Atlantic Region.

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REFERENCES

- ASSAD, L. T., D. S. GODMIN & M. OGAWA, 1996. Engorda de lagostas em viveiro do mar. Arquivos de Ciências do Mar, Fortaleza, **30**(1-2): 13-19.
- BAISRE, J. A., 1976. Distribución de las larvas de *Panulirus argus* y *Scyllarus americanus* (Crustacea, Decápoda) con notas sobre la abundancia y distribución de sus estadíos. Est. Inst. Oceanol., Cuba, **1**(1): 5-34.
- BAISRE, J. A., 1979. Reclutamiento de postlarvas y juveniles de langosta (*Panulirus argus* Latreille), en la plataforma suroriental. II Evento Científico de la Habana, Resumen Univ.: 1-102.
- BEVERTON, R. J. H. & S. J. HOLT, 1957. On the dynamics of the exploited populations. Fish. Invest., **19**: 1-533.
- CADDY, J. F., 1987. Size-frequency analysis for Crustacea: moult increment and frequency models for stock assessment. Kuwait Bulletin of Marine Science, **987**(9): 43-61.
- CAMPELL, N. A. & B. F. PHILLIPS, 1972. The Von Bertalanffy growth curve and its applications to capture — recapture data in fisheries biology. J. Cons. Int. Explor. Mer, Copenhagen, **34**(2): 295-299.
- CHANG, Y. J., CH. L. SUN, Y. CHEN & S. Z. YEH, 2012. Modelling the growth of crustacean species. Reviews in Fish Biology and Fisheries, **22**: 157-187.
- COBB, J. S. & J. F. CADDY, 1989. The population biology of decapods. In: J. F. CADDY (ed.), Marine invertebrate fisheries: 327-374. (J. Wiley and Sons, New York, NY).
- CRUZ, R. & R. BERTELSEN, 2009. The spiny lobster (*Panulirus argus*) in the Wider Caribbean: a review of life cycle dynamics and implications for responsible fisheries management. Proceedings of the Gulf and Caribbean Fisheries Institute, Annual Conference, **61**: 433-446.
- CRUZ, R., R. N. L. CONCEIÇÃO, R. A. MARINHO, J. C. BARROSO, J. S. HOLANDA, C. S. FÉLIX, M. E. O. MARTINS, F. S. SANTOS, K. C. A. SILVA & M. A. A. FURTADO-NETO, 2011. Metodologias de amostragem para avaliação das populações de lagosta: plataforma Continental do Brasil. Edição bilíngue português/espanhol, Fortaleza: UFC/LABOMAR/NAVE; UH/CIM, Coleção Habitat, **6**: 1-142.
- CRUZ, R., R. COYULA & A. T. RAMÍREZ, 1981. Crecimiento y mortalidad de la langosta espinosa (*Panulirus argus*) en la plataforma suroccidental de Cuba. Revista Cubana de Investigaciones Pesqueras, **6**(4): 89-119.
- CRUZ, R., R. COYULA & A. T. RAMÍREZ, 1986. Ecología de la langosta (*Panulirus argus*) al SE de la Isla de la Juventud. II. Patrones de movimiento. Revista de Investigaciones Marinas, **8**(3): 19-35.
- CRUZ, R., K. C. A. SILVA & I. H. A. CINTRA, 2013. Assessment of wild spiny lobster stocks on the Brazilian continental shelf. Crustaceana, **86**(3): 336-356.
- CRUZ, R., C. E. P. TEIXEIRA, M. O. B. MENEZES, J. V. M. SANTANA, T. M. NETO, J. C. GAETA, P. P. DE FREITAS, K. C. A. SILVA & I. H. A. CINTRA, 2015. Large-scale oceanic circulation and larval recruitment of the spiny lobster *Panulirus argus* (Latreille, 1804). Crustaceana, **88**(3): 298-323.
- DA COSTA, R. S. & D. LIMA PAIVA FILHO, 1974. Estudo da biologia da pesca de lagostas no Ceará — dados 1971-1973. Arquivos de Ciências do Mar, Fortaleza, **14**: 95-114.

- DAHLGREN, C. P. & F. STAINED, 2007. Growth and survival of Caribbean spiny lobster *Panulirus argus*, raised from puerulus to adult size in captivity. Proceedings of the Gulf and Caribbean Fisheries Institute, Annual Conference, **59**: 302-312.
- DO NASCIMENTO, I. V., I. B. TRAVASSOS, G. DE A. BORGES & M. E. ARAÚJO, 1984. Determinação dos parâmetros biométricos das lagostas *Panulirus argus* e *Panulirus laeviscauda* (Latr.) capturadas no litoral do Estado do Rio Grande do Norte. SUDENE, Recife: 1-31.
- DOS SANTOS, E. P. & C. T. C. IVO, 1973. Crescimento e idade da lagosta *Panulirus laeviscauda* (Latreille), em águas costeiras do Estado do Ceará (Brasil). Arquivos de Ciências do Mar, Fortaleza, **13**(1): 19-23.
- DOS SANTOS, E. P., R. D. C. SARAIVA & S. C. DE MOURA, 1964. Growth of the spiny lobster *Panulirus argus* (Latr.): quantitative aspect. Arquivos da Estação de Biologia Marinha da Universidade do Ceará, **4**(2): 41-44.
- EHRHARDT, N. M., 2008. Estimating growth of the Florida spiny lobster, *Panulirus argus*, from molt frequency and size increment data derived from tag and recapture experiments. Fisheries Research, **93**: 332-337.
- FAO, 1993. Yearbook of fishery statistics, catches and landings, **72**: 1-654. (FAO, Rome).
- FAO, 1996. Precautionary approach to capture fisheries and species introductions. E. Lysekil, Sweden, 6-13 June 1995. FAO Technical Guidelines for Responsible Fisheries, Rome, **2**: 1-54.
- FAO, 2006. Yearbooks of fishery statistics summary tables: 302-329. (FAO, Rome).
- FAO/WESTERN CENTRAL ATLANTIC FISHERY COMMISSION, 2001. Report on the FAO/DANIDA/CFRAMP/WECAFC regional workshops on the assessment of the Caribbean spiny lobster (*Panulirus argus*). Belize City, Belize, 21 April – 2 May 1997 and Merida, Yucatan, Mexico, 1-12 June 1998. FAO Fisheries Report, **619**: 1-381.
- FINGERMAN, S. W., 1985. Non-metal environmental pollutants and growth. In: A. M. WENNER (ed.), Factors in adult growth. Crustacean Issues, **3**: 219-234. (A. A. Balkema, Rotterdam).
- FONTELES-FILHO, A. A., 1992. Population dynamics of spiny lobsters (Crustacea: Palinuridae) in Northeast Brazil. Ciência e Cultura [Journal of the Brazilian Association for the Advancement of Science], **44**(2/3): 191-196.
- FONTELES-FILHO, A. A., 1994. A pesca predatória de lagostas no estado do Ceará: Causas e consequências. Boletim Técnico Científico, CEPENE, **1**(2): 107-131.
- FONTELES-FILHO, A. A. & A. F. HOLANDA, 1990. Análise do sistema de amostragem da captura de lagostas no Nordeste do Brasil. Caatinga, **7**: 175-185.
- FONTELES-FILHO, A. A., M. O. C. XIMENES & P. H. M. MONTEIRO, 1988. Sinopse de informações sobre as lagostas *Panulirus argus* e *Panulirus laeviscauda*, no Nordeste do Brasil. Arquivos de Ciências do Mar, **27**: 1-19.
- FORD, E., 1933. An account of the herring investigations conducted at Plymouth during the years from 1924 to 1933. Journal of the Marine Biological Association of the United Kingdom, **19**(1): 305-384.
- GAYANILO, F. C., JR., M. SORIONO & D. PAULY, 1989. A draft guide to complete ELEFAN. (International Centre for Living Aquatic Resources Management, Manila).
- GONZÁLEZ-CANO, J. M. & C. A. S. ROCHA, 1995. Problems on the estimation of growth parameters for the spiny lobster *Panulirus argus* in the Caribbean and Northeast Brazil. In: G. T. SAKAGAWA (ed.), Assessment methodologies and management. Proceedings of the first world fisheries congress, **5**: 147-159.
- GULLAND, J. A., 1969. Manual of methods for fish stock assessment. Part 1. Fish population analysis. FAO Manuals in Fisheries Science, **4**: 1-154.
- HARTNOLL, R. G., 1985. Growth, sexual maturity and reproductive output. In: A. M. WENNER (ed.), Factors in adult growth. Crustacean Issues, **3**: 101-128. (A. A. Balkema, Rotterdam).
- HARTNOLL, R. G., 1983. Strategies of crustacean growth. In: J. K. LOWRY (ed.), Papers from the conference on the biology and evolution of Crustacea. Australian Museum Memoir, **18**: 121-131.

- IVO, C. T. C. & J. A. PEREIRA, 1996. Crecimento e idade das lagostas *Panulirus argus* (Latreille), e *Panulirus laevicauda* (Latreille), no Norte e Nordeste do Brasil. Arquivos de Ciências do Mar, Fortaleza, **30**(1-2): 33-39.
- LEOCÁDIO, A. M. & R. CRUZ, 2008. Growth parameters of the spiny lobster *Panulirus argus* in the Great Caribbean: a review. Revista de Insuetigaciones Marinas, **29**(3): 239-248.
- LITTLE, E. J., 1972. Tagging of spiny lobsters (*Panulirus argus*) in the Florida Keys, 1967-1969. Florida Marine Research Publications, (Spec. Sci. Rep.) **31**: 1-23. (Florida Department of Natural Resources Marine Research Laboratory).
- LOZANO ALVAREZ, E., P. BRIONES FORUZAN & B. F. PHILLIPS, 1991. Fisheries characteristics growth, and movements of spiny lobster *Panulirus argus* in Bahía de la Asención, México. Fisheries Bulletin, **89**(1): 79-89.
- LYONS, W. G., D. G. BARBER, S. M. FOSTER, F. S. KENNEDY, JR. & G. R. MILANO, 1981. The spiny lobster, *Panulirus argus*, in the middle and upper Florida Keys: population structure, seasonal dynamics, and reproduction. Florida Marine Research Publications, **38**: 1-38.
- MATEO, I. & W. J. TOBIAS, 2002. Preliminary estimations of growth, mortality and yield per recruit for the spiny lobster *Panulirus argus*, in St. Croix, USVI. Proceedings of the Gulf and Caribbean Fisheries Institute, Annual Conference, **54**: 58-75.
- MATTHEWS, T. R. & K. E. MAXWELL, 2007. Growth and mortality of captive Caribbean spiny lobsters, *Panulirus argus*, in Florida, USA. Proceedings of the Gulf and Caribbean Fisheries Institute, Annual Conference, **58**: 358-366.
- MORGAN, M. D., 1985. Growth and its relationship to reproduction in *Mysis relicta*. In: A. M. WENNER (ed.), Factors in adult growth. Crustacean Issues, **3**: 235-250. (A. A. Balkema, Rotterdam).
- MORGAN, G. R., 1980. Populations dynamics of spiny lobsters. In: J. S. COBB & B. F. PHILLIPS (eds.), The biology and management of lobsters, **2**, ecology and management: 189-298. (Academic Press, New York, NY).
- O'BRIAN, J. & P. VAN WYK, 1985. Effects of crustacean parasitic castrators (epicaridean isopods and rhizocephalan barnacles) on growth of crustacean hosts. In: A. M. WENNER (ed.), Factors in adult growth. Crustacean Issues, **3**: 191-218. (A. A. Balkema, Rotterdam).
- OLSEN, D. & I. G. KOBLEC, 1975. Population dynamics ecology and behavior of spiny lobster *Panulirus argus* of St. John, Virgin Island, 2, growth and mortality. Natural History Museum of Los Angeles, Bulletin, **20**: 17-21.
- PAULY, D. & F. C. GAYANILO, JR., 1990. The growth of ELEFAN. Naga, the ICLARM Quarterly, **13**(2): 14-16.
- PAULY, D. & G. R. MORGAN (eds.), 1987. Length-based methods in fisheries research. ICLARM Conference Proceedings, **13**: 1-468. (International Center for Living Aquatic Resources Management, Manila, Philippines, and Kuwait Institute for Scientific Research, Safat, Kuwait).
- PEACOCK, N. A., 1974. A study of the spiny lobster fishery of Antigua and Barbuda. Proceedings of the Gulf and Caribbean Fisheries Institute, Annual Conference, **26**: 117-130.
- PHILLIPS, B. F., M. J. PALMOER, R. CRUZ & J. T. TRENDALL, 1992. Estimating growth of the spiny lobsters *Panulirus cygnus*, *Panulirus argus* and *Panulirus ornatus*. Aust. J. Mar. Freshw. Res., **43**: 1177-1188.
- PINTO PAIVA, M. & R. S. DA COSTA, 1965. Estudos de biologia da pesca de lagostas no Ceará — dados de 1964. Arquivos da Estação de Biologia Marinha da Universidade do Ceará, **5**(2): 127-150.
- PINTO PAIVA, M. & R. S. DA COSTA, 1966. Estudos de biologia da pesca de lagostas no Ceará — dados de 1965. Arquivos da Estação de Biologia Marinha da Universidade Federal do Ceará, **76**(2): 167-193.
- PINTO PAIVA, M. & R. S. DA COSTA, 1967. Estudos de biologia da pesca de lagostas no Ceará — dados de 1966. Arquivos da Estação de Biologia Marinha da Universidade Federal do Ceará, **7**(2): 147-171.

- PINTO PAIVA, M. & R. S. DA COSTA, 1968. Estudos de biologia da pesca de lagostas no Ceará — dados de 1967. Arquivos da Estação de Biologia Marinha da Universidade Federal do Ceará, **8**(2): 187-210.
- PINTO PAIVA, M. & R. S. DA COSTA, 1970. Estudos de biologia da pesca de lagostas no Ceará — dados de 1969. Arquivos de Ciências do Mar, Fortaleza, **10**(2): 131-142.
- PINTO PAIVA, M. & A. B. DA SILVA, 1962. Estudos de biologia da pesca de lagostas no Ceará — dados de 1961. Arquivos da Estação de Biologia Marinha da Universidade do Ceará, **2**(2): 21-34.
- RA'ANAN, Z. & D. COHEN, 1985. Ontogeny of social structure and population dynamics in the giant freshwater prawn, *Macrobrachium rosenbergii* (De Man). In: A. M. WENNER (ed.), Factors in adult growth. Crustacean Issues, **3**: 277-314. (A. A. Balkema, Rotterdam).
- RICKER, W. E., 1975. Computation and interpretation of biological statistics of fish populations. Bulletin of Fisheries Research Board of Canada, **191**: 1-382.
- ROCHA, C. A. S. & A. F. S. XAVIER, 2000. Relações biométricas das lagostas espinhosas *Panulirus argus* (Latreille) e *Panulirus laeviscauda* (Latreille) do Nordeste do Brasil. Arquivos de Ciências do Mar, Fortaleza, **33**: 179-182.
- ROLIM, A. E. & C. A. S. ROCHA, 1972. Biometria de lagostas jovens do gênero *Panulirus* Gray. Arquivos de Ciências do Mar, Fortaleza, **12**(2): 91-97.
- SCHWAMBOORN, R., T. K. MILDENBERGER & M. H. TAYLOR, 2019. Assessing sources of uncertainty in length-based estimates of body growth in populations of fishes and macroinvertebrates with bootstrapped ELEFAN R. Ecological Modelling, **393**: 37-51.
- SUTCLIFFE, W. H., 1957. Observations of the growth rate of the immature Bermuda spiny lobster *Panulirus argus*. Ecology, **38**: 526-529.
- TRAVIS, D. F., 1954. The molting cycle of the spiny lobster, *Panulirus argus* Latreille. I. Molting and growth in laboratory-maintained individuals. Biological Bulletin, Woods Hole, **107**(3): 433-450.
- VAUGHAN, D. S. & P. KANCIRUK, 1982. An empirical comparison of estimation procedures for the Von Bertalanffy growth equation. J. Cons. Int. Explor. Mer, **40**: 211-219.
- VON BERTALANFFY, L., 1938. A quantitative theory of organic growth (inquiries on growth laws, II). Hum. Biol., Baltimore, **10**(2): 181-213.
- WAHLE, R. A. & M. J. FOGARTY, 2006. Growth and development: understanding and modelling growth variability in lobsters. In: B. F. PHILLIPS (ed.), Lobsters: biology, management, aquaculture and fisheries: 1-43. (Blackwell Publishing Ltd., Oxford).
- WALFORD, L. A., 1946. A new graphic method of describing the growth of animals. Biological Bulletin, Woods Hole, **90**: 141-147.
- WAUGH, G. T., 1981. Management of juvenile spiny lobster *Panulirus argus* based on estimated biological parameters from Grand Bahama Island, Bahamas. Proceedings of the Gulf and Caribbean Fisheries Institute, Annual Conference, **33**: 271-289.

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